INCH-POUND

A-A-59381
19 November 2003

COMMERCIAL ITEM DESCRIPTION
OHMMETER

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This Commercial Item Description describes a soil resistance meter. The soil resistance meter is used to measure earth resistance and ground sensitivity. This CID is meant as a minimum requirement for the soil resistance meter, herein referred to as ohmmeter, in which only those manufacturers that meet or surpass the following requirements are supplied per this CID.

2. SALIENT CHARACTERISTICS. The ohmmeter shall be capable of operation within the accuracies, limits, and specifications herein. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be Class 3 in accordance with MIL-PRF-28800.

2.1 Safety. The equipment shall be UL listed (Underwriters Laboratory) or approved and/or shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

2.2 DC Power Sources. The ohmmeter shall operate from an internal DC power source. Maximum use will be made of standard, nomenclature battery-power sources. The use of nonstandard, non-nomenclature batteries requires the approval of the cognizant Air Force procuring activity. There shall be an external indication of the selected range.

2.3 Input Voltage Protection. If the ohmmeter operates at a regular frequency of 50/60/400 Hz ± 10%, it shall incorporate fuse protection to enable the ohmmeter to withstand nominal voltages not exceeding 1000 Vac or nominal voltages not exceeding 1500 Vdc into the test leads.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1611

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2.4 Battery State Indicator. A battery state indicator shall be provided to indicate when the capacity of batteries used for primary power is low. The approximate remaining operating time for the low battery state shall be specified in the operator's manual.

2.5 Battery Compartment. The ohmmeter shall provide a covered battery compartment. The compartment shall provide access only to the battery. The battery compartment shall provide protection from external sources that could damage the battery, protect the equipment from deteriorated batteries, and provide maintenance access to clean the compartment. The battery polarity, voltage, and type shall be marked in a prominent place on or adjacent to the battery compartment.

2.6 Battery Protection. The ohmmeter shall be protected against reversal of battery polarity.

2.7 Enclosure Requirement. The degree of protection provided by the enclosure shall allow the complete equipment to pass those environmental tests appropriate to Class 3 in accordance with MIL-PRF-28800.

2.8 Splash-Proof. The ohmmeter shall be splash-proof in accordance with MIL-PRF-28800.

2.9 Drip Proof. The ohmmeter shall be drip-proof in accordance with MIL-PRF-28800.

2.10 Soil Resistivity Test Kit. The soil resistivity test kit shall include all the equipment necessary to calculate the resistivity of the soil by the Wenner Four-Electrode Method, as outlined by the IEEE/ASTM G 57 standard. This kit shall include four-pin resistivity test reel with leads, four stainless steel T-handle pins 3/8" X 18" soil box with four soil pins, and a custom carrying case.

3.0 PERFORMANCE CHARACTERISTICS.

3.1 Warm Up Performance. The ohmmeter shall perform to specifications after a warm-up period not to exceed thirty minutes after initiating power and all calibration and self-test routines have been completed.

3.2 Resistance Range. The ohmmeter shall measure resistance from 0.01 ohm to 1.1 megohms.

3.3 Accuracy. The accuracy of the ohmmeter shall be at least ± 1% of full scale of any range. The resistance measurement of the ohmmeter shall be unaffected by the accessory test lead lengths, auxiliary electrode resistance, or stray AC or DC earth currents.

3.4 Maximum Depth. The ohmmeter and test kit shall be capable of a resultant resistivity measurement depth of twenty feet by use of the four-electrode method.

3.5 Measurement Method. The ohmmeter shall be capable of two, three or four terminal measurement. The null balance principle shall be applied to eliminate probe resistance from measurement.

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3.6 Calibration. The ohmmeter shall have a minimum calibration interval of twelve months. At the end of this interval, a minimum of 83% of the units shall remain in tolerance. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components that are adjustable by the use of simple means. The calibration of substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removable of the instrument case or modules.

3.7 Weight and Dimensions. The weight & dimensions of the ohmmeter shall be in accordance with standard industry practices.

3.8 Manuals. The ohmmeter shall be delivered with operator, maintenance, and illustrated parts manuals. Format and quantity shall be as specified in the contract or order.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). These are consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements of exposing the supplier's employees to undue hazards from the recovered materials.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of ASTM-SI10, and all other requirements of this CID are met.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order. The Equipment shall have an identification plate permanently affixed in readable place, which includes the information contained in MIL-PRF-28800, Section 3.

7. NOTES.

7.1 Sources of documents.

7.1.1 Government Documents. Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

7.1.2 Code of Federal Regulations (CRF). Copies of the CFR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

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7.1.3 ASTM Standards. Copies of ASTM standards can be obtained from ASTM, 100 Bar Harbor Drive, West Conshohocken, PA 19428-2959. Phone 610-832-9500. Fax 610-832-9555.

7.1.4 IEEE Standards. IEEE Standards may be obtained by contacting IEEE Operations Center, 445 Hoes Lane, PO Box 1331, Piscataway, NY, 08854-1331. Phone 800-678-4333, (Toll-Free, US & Canada), 732-981-0060, Fax: 732-981-9667 www.ieee.org.

MILITARY INTEREST

Custodian:
Air Force – 99

Preparing activity:
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